

Course information and policies

Aircraft Propulsion Systems

Lecturer

- M. Sc. Christian Simental – Thermal Power & Aerospace Propulsion
 - All enquiries through the class email:
 - csimentale@uanl.edu.mx
 - Email subject: Propulsion – Semester – Subject
- University email is going to be used for formal communication using the premade distribution lists for Teams. Such communication include and it's not limited to:
 - Exams
 - Absences
 - Assignments & Projects
- Review your status on teams/distribution list to not miss any message!!!
- Constantly review your university email. Make it a habit!!!

Overall course grade and attendance

- Course lectures are established as in person. Unless indicated by the lecturer.
 - Attendance will not be considered for the final grade.
- The final grade will be based in your performance in the following activities:
 - Failure to submit project represents a NC as final grade.

Evaluation	Weight
A1 - PIA intro report	5%
A2 - PIA ppt 1	15%
A3 - PIA report 1	30%
A4 - Mid Term Exam	5%
A5 - Final Exam	5%
A6 - PIA final presentation	15%
A7 - PIA final report	25%
Final Grade	100%

Course Objectives



Gas Turbine Theory and Performance

- Perform air-standard analyses of gas turbine power plants based on the Brayton cycle and its modifications, including:
 - sketching T-s diagrams and evaluating property data at principal states.
 - applying mass, energy, entropy, and exergy balances.
 - determining net power output, thermal efficiency, back work ratio, and the effects of compressor pressure ratio.
- For subsonic and supersonic flows through nozzles and diffusers:
 - demonstrate understanding of the effects of area change, the effects of back pressure on mass flow rate, and the occurrence of choking and normal shocks.
 - analyze the flow of ideal gases with constant specific heats.

Main Component Design

- discuss the main requirements for design of gas turbine components, different approaches and introduction to turbomachinery specific concepts.

Midterm and final exam

Dates

- Per academic calendar unless otherwise specified by the lecturer.
- All exams are to be taken in person.

What will it cover?

- All material reviewed in class, assignments and reading material prior the exam date.
- Final exam will cover all material reviewed after the midterm.

Grading and lecturer's expectations

- Expect you to answer with legible handwriting in a structured, complete and clean manner.
- The calculation method will be revised but doesn't mean you will get points weather the answer is right or wrong

PIA Generals

With no analysis the most you can get is 40 per report!

Teams

- 4 - 5 members
- Submittal of your team deadline by the second week of regular classes

Project 1st Deliverable

- Literature Review
- Aims and Objectives
- Technical Aspects
- Methodology and Approach

Presentations

- Proficiency on the subject
- Knowledge and familiarity with the work done.
- Handling the questions

Reports

- Work, Efforts, Results - **30%**
- Analysis and conclusions - **60%**
- Style, presentation and reporting - **10%**

Some expectations and key points on your reports in no particular order

Your report is not a history lesson.

Make it technology focused.

A graph/calculation/plot etc. is not the end. It is the start of the analysis.

Give time to enable the post processing and analysis of your results ... then double it!

Develop a clear argument on what is the "outcome" of your work. This is not a summary of what you have done.

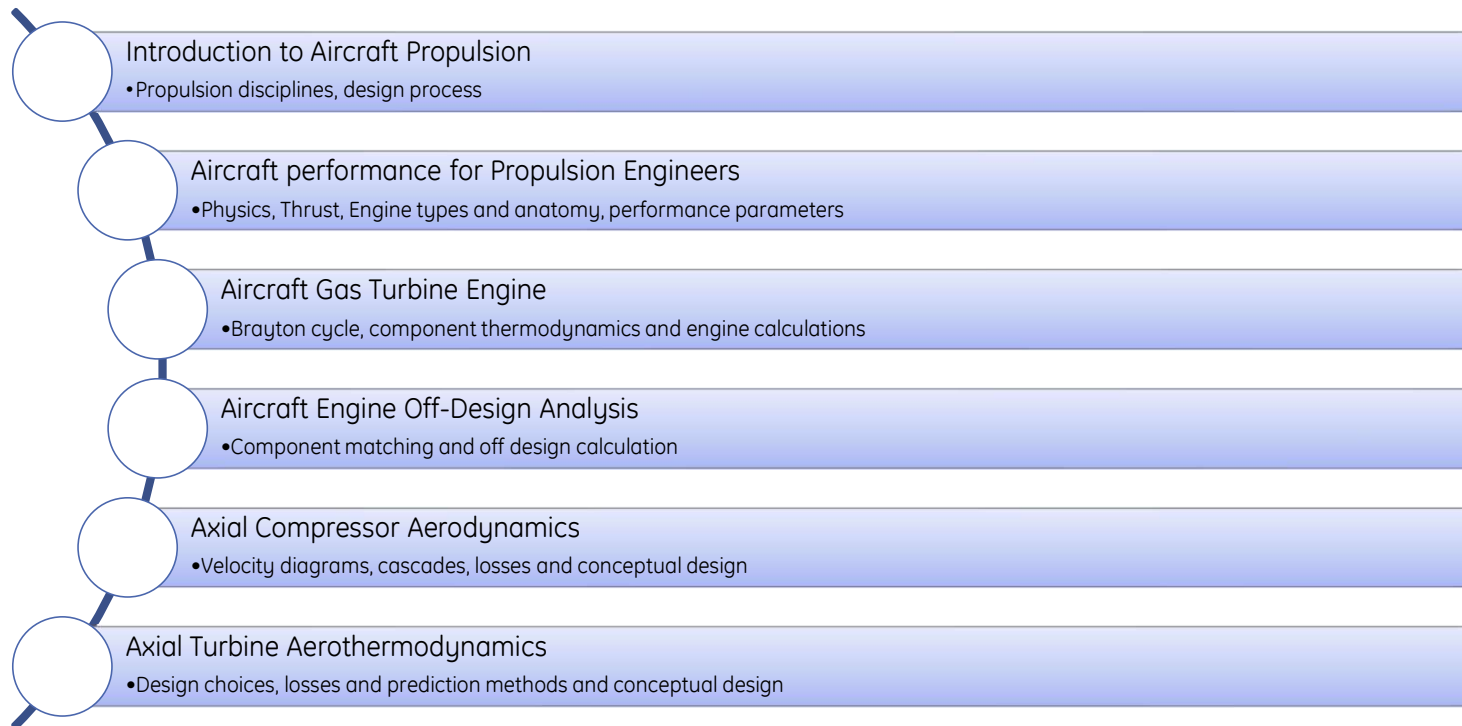
Be aware of the "SO WHAT?" factor when analyzing your work.

Be self critical.

Highlight your findings and create links between your work and past studies.

Comment on conflicts or agreements.

Syllabus



Planning and Deadlines

Presentations will be presented on the corresponding date but need to be submitted by the previous Monday in order to be considered.

Group 004	Group 002		Ppt
7-Aug	8-Aug		Class 1
14-Aug	15-Aug	Submit team members	Class2
21-Aug	22-Aug		Class3
28-Aug	29-Aug		Class3
4-Sep	5-Sep	A1 - PIA intro report	Class3
11-Sep	12-Sep		Class4
18-Sep	19-Sep		Class4
9-Oct	10-Oct	A2 - PIA ppt 1	
16-Oct	17-Oct		Class5
23-Oct	24-Oct	A3 - PIA report 1	Class5
30-Oct	31-Oct		Class5
6-Nov	7-Nov		Class6
13-Nov	14-Nov		Class6
20-Nov	21-Nov	A6 - PIA final presentation	
27-Nov	28-Nov	A7 - PIA final report	

Recommended literature

- Farokhi, S. (2014). *Aircraft Propulsion*. John Wiley & Sons.
- Mattingly, J. D., Boyer, K. M., & Ohain, H. (2016). *Elements of Propulsion: Gas Turbine and Rockets*. AIAA Education Series.
- Mattingly, J. D., Boyer, K. M., Haven, B. A., Heiser, W. H., & Pratt, D. T. (2019). *Aircraft Engine Design*. AIAA Education Series.
- Saravanamuttoo, H., Crichton Rogers, G., & Cohen, H. (2001). *Gas Turbine Theory*. Prentice Hall.
- Dixon, S., & Hall, C. (2010). *Fluid Mechanics and Thermodynamics of Turbomachinery*. Butterworth - Heinemann.

